

Analyzing Discrepancies in Human Vulnerability Metrics for Compound Weather Extremes

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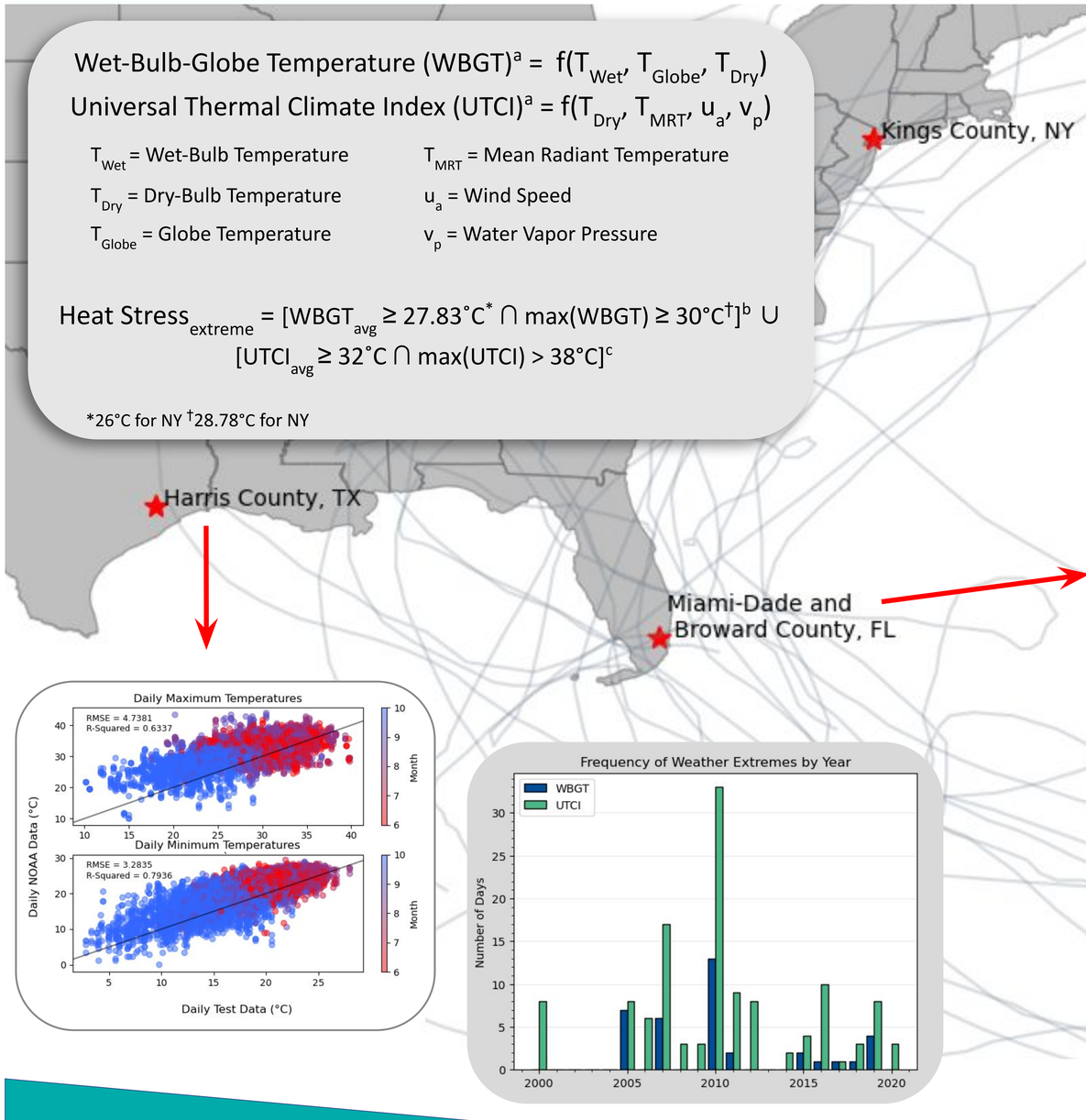
Wet-Bulb-Globe Temperature (WBGT)[a] = $f(T_{Wet}, T_{Globe}, T_{Dry})$

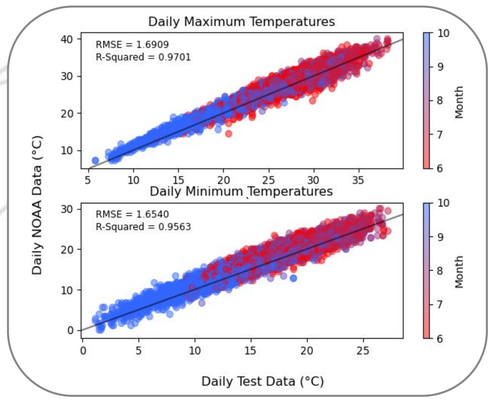
Universal Thermal Climate Index (UTCI)[a] = $f(T_{Dry}, T_{MRT}, u_a, v_p)$

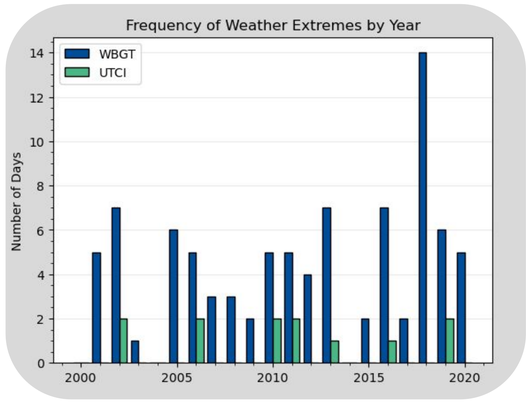
T_{Wet} = Wet-Bulb Temperature
T_{Dry} = Dry-Bulb Temperature
T_{Globe} = Globe Temperature
T_{MRT} = Mean Radiant Temperature
u_a = Wind Speed
v_p = Water Vapor Pressure

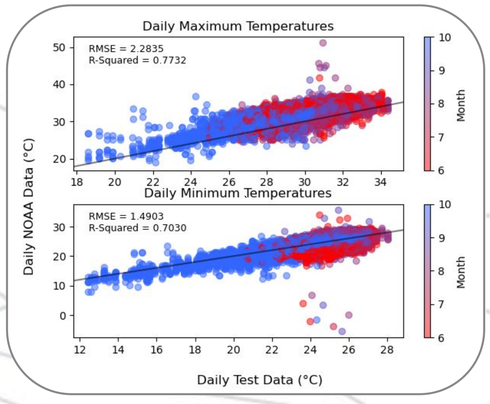
$$\text{Heat Stress}_{extreme} = [WBGT_{avg} \geq 27.83^{\circ}C^{*} \cap \max(WBGT) \geq 30^{\circ}C^{\dagger}]^{b} \cup [UTCI_{avg} \geq 32^{\circ}C \cap \max(UTCI) > 38^{\circ}C]^{c}$$

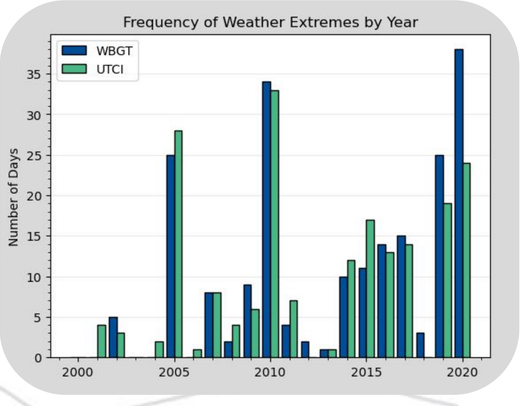
*26°C for NY †28.78°C for NY











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[a] Spangler, K. R., Liang, S., & Wellenius, G. A. (2022). Wet-bulb globe temperature, universal thermal climate index, and other heat metrics for US Counties, 2000–2020. Scientific data, 9(1),326. [b] Grundstein, A., Williams, C., Phan, M., & Cooper, E. (2015). Regional heat safety thresholds for athletics in the contiguous United States. Applied geography, 56, 55-60.; [c]Adapt, C.(2020). Thermal Comfort Indices—Universal Thermal Climate Index, 1979–2020. The European Climate Adaptation Platform Climate-ADAPT. The European Commission and the European Environment Agency).